

Riemann Sums and Definite Integrals

ANSWER KEY



Approximating area with Riemann sums

- 1 Estimate $\int_0^4 x^2 dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.
 $\Delta x = 1$; right endpoints 1, 2, 3, 4: sum $= 1(1) + 1(4) + 1(9) + 1(16) = 30$.
- 2 Estimate $\int_1^5 (2x + 1) dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.
 $\Delta x = 1$; left endpoints 1, 2, 3, 4: values 3, 5, 7, 9: sum $= 24$.

The definite integral and its properties

- 3 Given $\int_0^3 f(x) dx = 8$ and $\int_3^7 f(x) dx = -2$, find:
a $\int_0^7 f(x) dx = 6$ b $\int_7^3 f(x) dx = 2$ c $\int_0^3 5f(x) dx = 40$
- 4 The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x-axis on $[0, 2]$.
Evaluate $\int_0^2 (4 - x^2) dx$ exactly.
 $\left[4x - \frac{x^3}{3}\right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}$.
- 5 Evaluate $\int_{-2}^2 (x^3 - 4x) dx$ by recognizing the integrand's symmetry, without computing an antiderivative.
The integrand is odd ($f(-x) = -f(x)$), and the interval is symmetric about 0, so the integral is 0.